

# **HedgeMind: The Next Evolution in Tactical Hedge Fund Operations**

End Evaluation Report

Team 76



Inter IIT Tech Meet 14.0



# 1 Introduction

This project presents a Real-Time AI Copilot for Tactical Hedge Fund Managers, a specialized multi-agent system designed to bridge the gap between high-velocity market data and institutional-grade decision-making. As financial markets become increasingly volatile and data-dense, the cognitive load on managers to simultaneously analyze technical indicators, macroeconomic shifts, and regulatory filings has reached critical mass. Our objective is to alleviate this bottleneck by deploying an autonomous but compliant, streaming-native architecture that serves not just as a search tool, but as a sophisticated reasoning engine capable of complex financial workflows.

## 1.1 Novelty

Our research into industry-standard tools, such as **Citadel's** AI Assistant and the **Bridgewater Associate's** AIA Forecaster, reveals a landscape of impressive *strategic* capabilities but critical *tactical* vulnerabilities. While these SOTA systems excel at compressing hours of deep-dive equity research into minutes, they fundamentally rely on probabilistic LLMs for decision-making. In a Tactical Hedge Fund (THF) environment-where alpha depends on exploiting fleeting arbitrage in futures markets or reacting instantly to central bank shifts-relying on probabilistic inference without deterministic guarantees poses an existential threat. The latency inherent in iterative "thought loops" renders them unsuitable for high-velocity execution where precision is non-negotiable.

Our solution addresses these failure modes through a unique architecture designed specifically for the leveraged, derivative-heavy environment of tactical trading:

- **Streaming-Native Execution via Trigger Agents:** Competitor models rely on high-latency search-and-reason loops. This is fatal for CTAs where data velocity outpaces processing rates. By leveraging **Pathway**, our system utilizes a **push-based streaming architecture** that maintains live materialized views of the market state. We employ a dedicated **Trigger Agent** that continuously monitors high-frequency signals-including **volume spikes**, **market sentiment shifts**, and **trend confidence levels**. This ensures that decisions regarding interest rate swaps or commodity futures are made on *live* ticks.
- **Deterministic Leverage Control:** In THFs, where leverage is standard, an LLM hallucination is a potential margin call. We solve this by decoupling reasoning from execution using **deterministic tool components**. We ingest specific **hedge fund-given constraints** on trade parameters to solve **constrained optimization** problems. This ensures that even if the AI suggests a high-conviction macro bet, the system strictly adheres to company policies and risk mandates.
- **Verifiable Macro Intelligence:** Global Macro funds are uniquely vulnerable to both external adversarial data poisoning and internal information leakage. Unlike standard tools that blindly ingest and expose top-ranked search results, our system implements a **CRAG** which actively verifies the epistemic validity of regulatory filings and macro indicators before they enter the decision stream.
- **Scalable Stateless Agents:** Tactical strategies often require rebalancing hundreds of positions simultaneously across uncorrelated assets. Standard multi-agent architectures suffer from non-deterministic latency when scaled. We employ a unique architectural split: **Stateless Domain Agents** controlled by a **Stateful Orchestrator**. This allows our system to handle the massive concurrent throughput required by high-frequency rebalancing algorithms without timeout failures.

## 1.2 Problem Understanding & Use-Case Validation

The FinTech landscape is pivoting to **Autonomous Hypothesis Generation** (e.g., Man Group's AlphaGPT), where the bottleneck is synthesizing data, not just acquiring it. However, deploying **Black-Box** traders remains institutionally non-viable due to strict risk regulations.

- **Industry Validation:** Our research confirms a rigid dichotomy: firms demand AI power but require compliance oversight. **Julia Liu (Head of Finance, fal.ai; Ex-Goldman Sachs)** validates our decoupled architecture, noting that banks "*fear AI being inexplicably wrong*" and that our deterministic constraints mimic industry-standard "**Maker-Checker**" protocols. Similarly, **Charudatt Mulye (Head of Equity & Data Science, India)** emphasized that standard alerts are commodities; our integration of **News Sentiment** acts as a "*differentiator*" by providing the "why" behind price moves.
- **The Hedge Fund Use Case:** Focusing on Tactical Funds, **Rupal Shah (Analyst, Goldman Sachs)** highlighted that minimizing "churn" is critical. She validated our **Trigger Agent's hysteresis** as effectively coding "*trader discipline*" into the bot, and described our hard-coded risk limits as "*liberating*," allowing managers to trust the system.

To meet this standard, the Assistant must operate in real-time. Standard database architectures introduce fatal latency. This



is where **Pathway** becomes the definitive enabler. By maintaining live indexes and performing stateful streaming, Pathway democratizes institutional-grade capability—offering the throughput of Kafka with the depth of SQL—granting managers immediate intelligence without the lag of traditional systems. For a more elaborate transcript of these industry feedback interviews, please refer to the Appendix.

### 1.3 Solution Overview

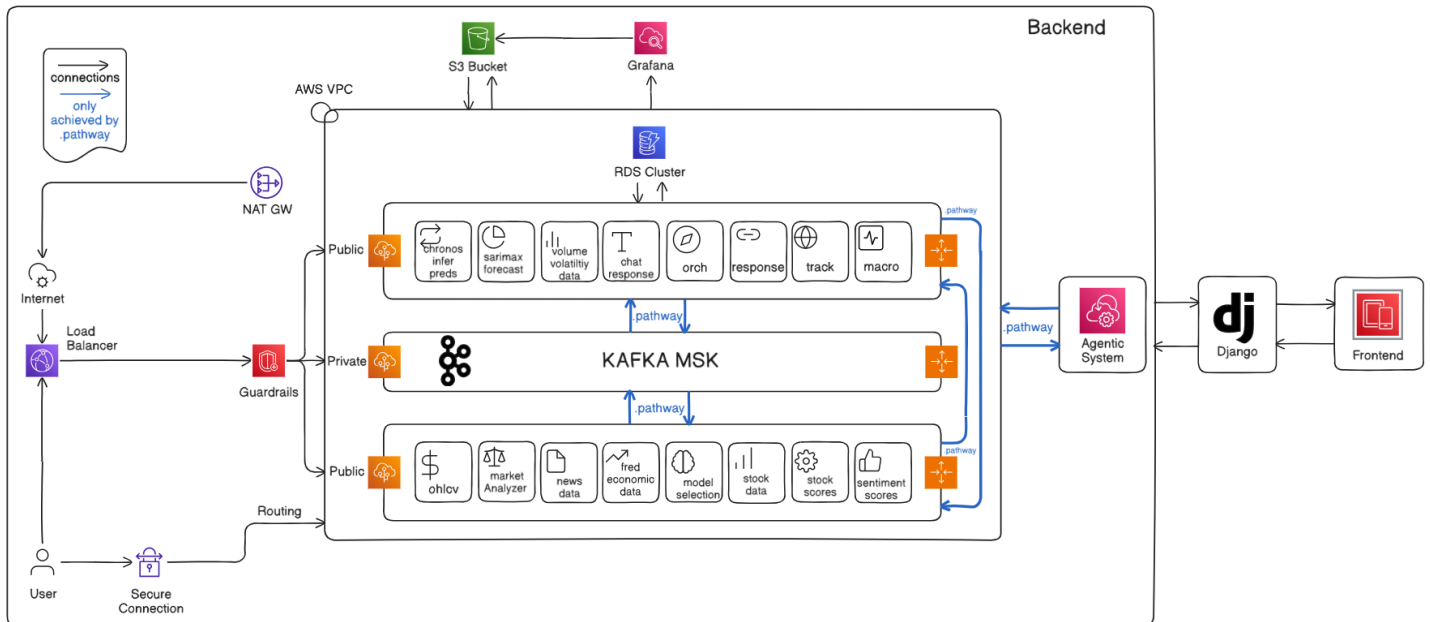


Figure 1: System Architecture Overview

Targeting 100 US large-cap stocks across 11 GICS sectors, our solution utilizes a hierarchical pipeline orchestrated via the Model Context Protocol (MCP). Following Clarification Agent validation, an Orchestrator routes tasks to specialized domain agents (Market, Macro, Portfolio, Strategy). The Market and Macro agents employ dedicated Corrective RAG instances with check-and-fetch logic—verifying internal vectors (technical signals, regulatory filings) before triggering Web Search fallback—to ensure high data fidelity. Simultaneously, a Trigger Agent optimizes predictive accuracy by dynamically switching between Chronos-2 and SARIMAX models based on live performance metrics and sends notifications in the events of extreme change.

**Advisory Research Assistant Role:** Explicitly functioning as an Advisory Research Assistant, the system provides reasoning and signal generation but lacks autonomous execution authority. Final trade decisions are bounded by deterministic logic, ensuring the AI acts as an intelligent copilot within a strict "human-in-the-loop" compliance framework.

## 2 System Architecture

McKinsey and CWAN highlight the necessity of a **Project Manager orchestrator** overseeing specialized stateless agents to achieve auditable scalability. Adhering to this principle, our core workload is distributed across specialized, LLM-driven agents (Market, Macro, Portfolio, Strategy) that operate in parallel.

### 2.1 The Data Layer: Tiered Storage Architecture

We complement Pathway's high-velocity in-memory processing with a tiered storage architecture to ensure durability and auditability without compromising real-time latency.

- **Structured Transactional State (PostgreSQL):** Persists the Dynamic Window metadata generated by Pathway. This allows Pathway to focus on complex, stateful windowing in-memory, while PostgreSQL ensures transient window states are durably recorded for recovery and audit trails.
- **Historical Market Data (S3/Parquet):** Offloads high-volume tick data and sentiment logs to Parquet files. This supports deep-regime backtesting for the Strategy Agent without bloating the live inference stream.
- **Vector Persistence:** Utilizes **Pathway's** native persistent vector store to maintain RAG embedding contexts (e.g., SEC filings) across system restarts, ensuring reasoning continuity.

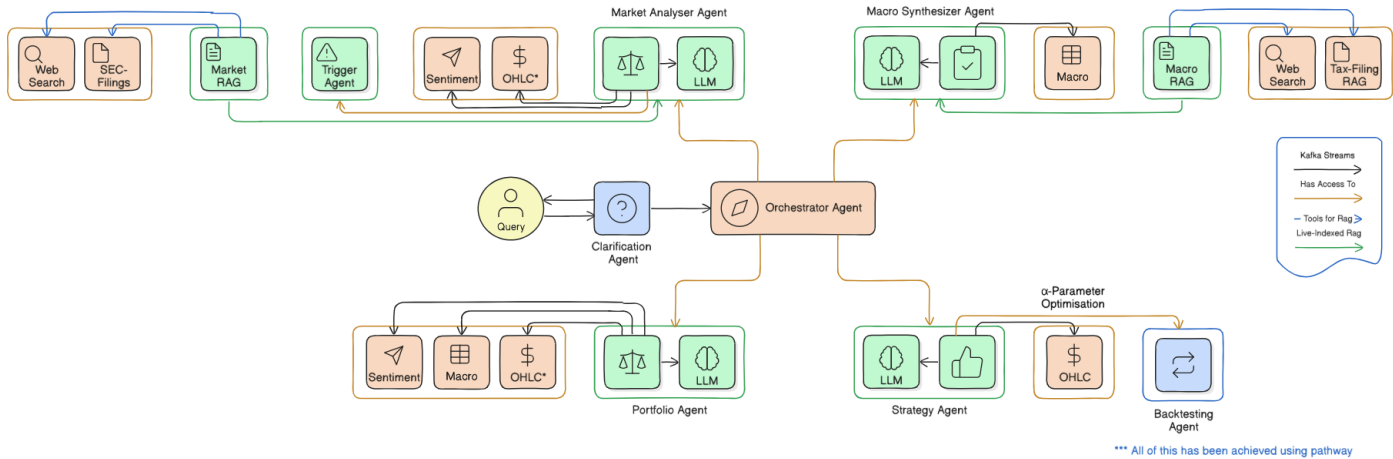


Figure 2: System Architecture Overview

## 2.2 The Safeguard Layer: Clarification Agent

Aligning with Digiqt and Alinia on the necessity of deterministic validators to prevent market manipulation or PII extraction, our pipeline enforces operational integrity via a mandatory Clarification Agent.

- **Guardrail Enforcement:** Utilizing a specialized **GPT-4o safeguard model**, it pre-screens all inputs, strictly rejecting any requests that violate safety guardrails or suggest unethical trading practices.
- **Intent Resolution:** Beyond safety, it validates financial intent. If instructions are ambiguous (e.g., rebalancing without constraints), the agent initiates a multi-turn dialogue to resolve missing parameters, preventing execution on vague inputs.

## 2.3 Trigger Agent

Overcoming passive limitations, the Trigger Agent operates independently to detect high-confidence anomalies through autonomous surveillance.

- **Multi-Modal Surveillance:** The agent continuously scans high-frequency streams via **Pathway**, monitoring market microstructure (liquidity, volume) and news sentiment.
- **Dual Confidence Validation:** To filter noise, it enforces **Dual-Check Logic**: a trigger requires predictive confidence to exceed a high threshold and persist over a specific hysteresis window, ensuring reaction only to genuine trend formations.
- **Dynamic Model Switching:** Acting as a predictive arbiter, it monitors Hybrid Forecasting Engine residuals. If **SARIMAX** linear assumptions degrade, the agent instantly hot-swaps to **Chronos-2** to capture non-linear volatility.
- **Scalable Infrastructure:** Heavy computation is decoupled from decision logic. The concurrent training for the 100-stock universe runs on a **Kubernetes** cluster, ensuring resource-intensive processing never bottlenecks the real-time loop.
- **Instant Action:** Upon validation, the agent simultaneously pushes a context-rich event to the pipeline and issues a high-priority system alert, ensuring immediate visibility of actionable volatility.

## 2.4 The Control Layer: Stateful Orchestration

While Agentuity systems rely on persistent context for workflow continuity, our Orchestrator Agent advances this by functioning as a "Stateful Brain" within the MCP framework.

- **Deep Context:** Unlike stateless routers, the Orchestrator maintains deep session memory and full conversation history, enabling it to execute compounded logic across multi-step financial tasks.
- **Strategic Delegation:** Acting strictly as a manager, it dynamically routes tasks to domain agents based on validated intent. This ensures the system remains goal-oriented, delegating execution to the appropriate tool while maintaining high-level oversight.

## 2.5 The Execution Layer: Stateless Domain Agents

Aligning with McKinsey and CWAN's Project Manager pattern for auditable scalability, our core workload is distributed across specialized, stateless agents. These units—acting as LLM controllers for the aforementioned agents operate in parallel to maximize yield.



### 2.5.1 Market Analyser Agent

Acting as the bridge between raw ticks and tactical intelligence, this agent operates through a multi-layered data pipeline:

- **Real-Time Data Access:** Using direct access to OHLCV streaming tables via Pathway, the agent ingests continuous events (volume spikes, breakouts) to synthesize real-time market updates and end-of-day summaries.
- **Dual Predictive Engine:** To provide forward-looking insights, it interfaces with the Trigger Agent, automatically querying the most statistically robust forecast from the selected model (Chronos-2 or SARIMAX).

**Dedicated Retrieval Systems:** The agent employs a specialized retrieval stack: a private **CRAG** indexed on **SEC Filings** (10-Ks, 10-Qs) to correlate price action with disclosed risks, and a restricted **Web Search Tool** to verify live context (breaking news, rumors) from high-authority domains.

### 2.5.2 Macro Agent

Conducting top-down market analysis, this agent fuses quantitative data with regulatory insights. It first employs a Macro Tool to ingest economic indicators (CPI, PPI, Interest Rates) via the FRED API, generating real-time sector index rankings. These signals are subsequently validated against a specialized CRAG indexed exclusively on tax and regulatory filings.

- **CRAG-based Verification:** Using strict verification mechanisms, the agent cross-references flagged sectors against recent tax code change disclosures retrieved from the RAG.
- **Self-Correction:** The system evaluates semantic relevance of retrieved documents. If search confidence is low, it triggers a corrective web search fallback for high-authority regulatory news, preventing hallucinations based on obsolete tax data.

### 2.5.3 Portfolio Agent

Acting as the system's "Quantitative Engine," this agent eliminates static model latency through two key mechanisms:

- **Dynamic Risk Modeling:** Replacing legacy batch processing—which often misses black swan events or flash crashes—this agent ingests continuous multi-modal streams (stock scores, real-time sentiment, macro sector rankings). This facilitates immediate, event-driven risk assessment, allowing the fund to react to volatility instantly rather than on rigid, calendar-based schedules.
- **Constraint-Based Optimization:** Covering Portfolio Creation, Optimization, and Diversification, the agent utilizes rigorous solvers (Mean-Variance, CVaR, Omega strategies) rather than opaque black boxes. These models are dynamically recalibrated on live data to detect risk clusters and are strictly bounded by institutional constraints (Beta targets, Sharpe thresholds, Sector Exposure limits) to ensure production-grade, operationally valid outputs.

### 2.5.4 Strategy Agent

The Strategy Agent leverages advanced evolutionary computing for alpha generation, surpassing traditional optimization limitations.

- **Evolutionary Discovery via STVGP:** Transitioning from scalar Genetic Programming (GP) to Strongly-Typed Vectorial GP (STVGP), the model processes technical indicators (e.g., Moving Averages, RSI) as input vectors with rolling windows rather than isolated scalar points. This approach captures essential temporal context and multi-day correlations, delivering superior profitability and stability compared to scalar variants. Furthermore, strict boolean constraints ensure the evolution of human-readable trading rules.
- **Backtesting Engine:** This robust module validates both internally evolved and user-uploaded (CSV) strategies. It employs historical simulations and walk-forward validation to verify performance and robustness prior to deployment.

## 3 Results and Metrics

We validated the system through component-level testing of forecasting and verification modules, followed by a comprehensive audit of portfolio strategies against standard benchmarks.

### 3.1 Component Validation

Prior to trade execution, individual modules were assessed for predictive accuracy and reliability.

#### 3.1.1 Price Forecasting (Hybrid Ensemble)

We evaluated the predictive power of our underlying forecasting engines -combined to form our hybrid Pathway Ensemble which dynamically switches between them based on regime residuals

**Table 1:** Forecasting Engine Performance (RMSE & Accuracy)

| Model            | RMSE  | MAE  | MAPE   | Directional Accuracy | Volatility Capture |
|------------------|-------|------|--------|----------------------|--------------------|
| Pathway Ensemble | 8.088 | 2.99 | 76.50% | 70.78%               | Dynamic            |

As shown in Table 1, the model significantly minimized error for the financial year 2021-22

### 3.1.2 Regime Detection & Verification

- **Macro-Economic Forecasting (LSTM):** The quarterly regime-detection model demonstrated robust predictive capability (Table 2).
- **Unstructured Data Verification (RAG LLM):** Acting as a "Judge" agent to filter hallucinated signals and adversarial noise, the RAG module achieved a **Validation Accuracy of 92%**, ensuring high-fidelity trigger generation.

**Table 2:** Macro LSTM Performance Metrics (Quarterly Horizon)

| Model      | MSE    | MAPE   | MAE    | Directional Accuracy |
|------------|--------|--------|--------|----------------------|
| Macro LSTM | 0.0073 | 0.243% | 0.0097 | 57.10%               |

## 3.2 Portfolio Performance Audit

We benchmarked our three core services-**Creator** (Alpha), **Rebalancer** (Risk Control), and **Diversifier** (Omega)-against the S&P 500 (SPY) during a high-volatility regime for the financial year 2021-22

**Table 3:** Portfolio Services vs. S&P 500 Baseline (2021–2022)

| Service                       | Total Return   | Sharpe Ratio | Volatility | Max Drawdown   | Beta |
|-------------------------------|----------------|--------------|------------|----------------|------|
| <i>S&amp;P 500 (Baseline)</i> | -18.22%        | -0.77        | 21.35%     | -25.43%        | 1.00 |
| <b>Rebalancer</b>             | <b>9.61%</b>   | <b>0.45</b>  | 62.41%     | <b>-14.20%</b> | 1.88 |
| <b>Diversifier</b>            | <b>-16.28%</b> | <b>-0.16</b> | 43.79%     | <b>-19.80%</b> | 1.77 |
| <b>Creator</b>                | -19.75%        | <b>-0.36</b> | 37.33%     | <b>-21.50%</b> | 1.59 |

## 3.3 Strategy Agent Analysis

To isolate the efficacy of the Genetic Programming (GP) Strategy Agent, we compared its performance for the financial year 2021-22 against a Buy-and-Hold (B&H) strategy on assets with distinct market behaviors: **Tesla (TSLA)** (momentum) and **Blackstone (BX)** (drawdown).

**Table 4:** GP Strategy Agent vs. Buy-and-Hold Performance

| Asset       | Strategy          | ROI           | Sharpe      | Max Drawdown   |
|-------------|-------------------|---------------|-------------|----------------|
| <b>TSLA</b> | <b>GP (Agent)</b> | <b>30.58%</b> | <b>2.70</b> | <b>-20.31%</b> |
|             | B&H               | 8.30%         | 0.31        | -50.44%        |
| <b>BX</b>   | <b>GP (Agent)</b> | <b>3.02%</b>  | <b>3.24</b> | <b>-1.99%</b>  |
|             | B&H               | -30.34%       | -0.33       | -43.57%        |

# 4 Implementation Details & Production Bar

## 4.1 System Architecture: Streaming Ingest & Live Indexing

- **Real-Time Data Ingestion:** The data pipeline initiates with the ingestion of live market data from external APIs, streamed directly into Apache Kafka topics. Downstream consumption is handled via Pathway Kafka connectors, enabling the low-latency integration of continuous data streams into the processing logic.
- **Agent-to-Agent Communication Strategy:** Inter-agent communication is architected as a streaming, asynchronous system. We utilize a custom Agent-to-Agent (A2A) protocol over Kafka, enforcing strict schema validation to ensure data integrity between components.



- **State Management & User Persistence:** To maximize horizontal scalability, the architecture enforces a strict separation of concerns regarding state:
  - **Stateless Agents:** Computational agents are designed to be entirely stateless. Context is injected dynamically by the orchestrator at the query level, removing the need for local memory persistence within individual agents.
  - **Stateful Orchestrator with Branching Logic:** The central orchestrator manages conversation state using a PostgreSQL backend. Uniquely, this state handling implements a "History Truncation" protocol: if a user edits a previous input, the Orchestrator automatically truncates the stored context to that specific timestamp. This allows the conversation to "fork" or restart from that exact interaction point without context contamination from the abandoned thread.
- **RAG & Specialized Indexing:** The Retrieval-Augmented Generation (RAG) agents leverage live indexing to maintain up-to-date context. We implemented a novel custom chunking algorithm specifically optimized for SEC filings, addressing the complexity of high-density tabular data often found in financial reports to ensure precise quantitative retrieval.

## 4.2 Production Practices: Resilience, Observability, & Guardrails

The production environment is designed for operational maturity, emphasizing fault tolerance, security, and deployment standardization.

- **Tool Integration & MCP Compliance:** Agent capabilities are extended through a modified implementation of the Pathway LLM xPack.
  - **Internal Tools:** Tools outputting directly from Pathway utilize the Pathway **Model Context Protocol (MCP)** servable class.
  - **External Services:** Third-party integrations are hosted on distributed servers and accessed via the standard MCP protocol, ensuring a unified interface for tool execution. This allows for seamless integration via API to existing OMS/EMS systems standard in hedge funds.
- **Resilience & Computational Persistence:** System reliability is bolstered by a multi-layered error handling strategy.
  - **Pathway Native Persistence:** We leverage Pathway's **internal persistence engine** to guarantee pipeline integrity. If any computation node or agent fails mid-process, the system does not need to replay the entire stream. Instead, it restarts the computation graph exactly from the last successful checkpoint, ensuring zero data loss and seamless recovery of stateful windowing operations.
  - **Graceful Degradation:** Failures trigger verbose error messaging combined with exponential backoff algorithms for rate-limit management. Agents are equipped with explicit meta-instructions on how to interpret specific tool failures, ensuring the pipeline does not stall on transient errors.
- **Security & Guardrails:**
  - **Adversarial Filtering:** A dedicated Clarification Agent acts as a firewall against prompt injection attacks, preventing unauthorized access to system prompts or proprietary datasets.
  - **Domain Restriction:** To mitigate hallucinations, the web search tool is constrained to a strict whitelist of trusted financial domains.
- **Deployment & CI/CD:** The entire pipeline is containerized using Docker to ensure environment consistency and is deployed on AWS infrastructure. This setup supports a simplified setup flow and reproducible builds across development and production stages.

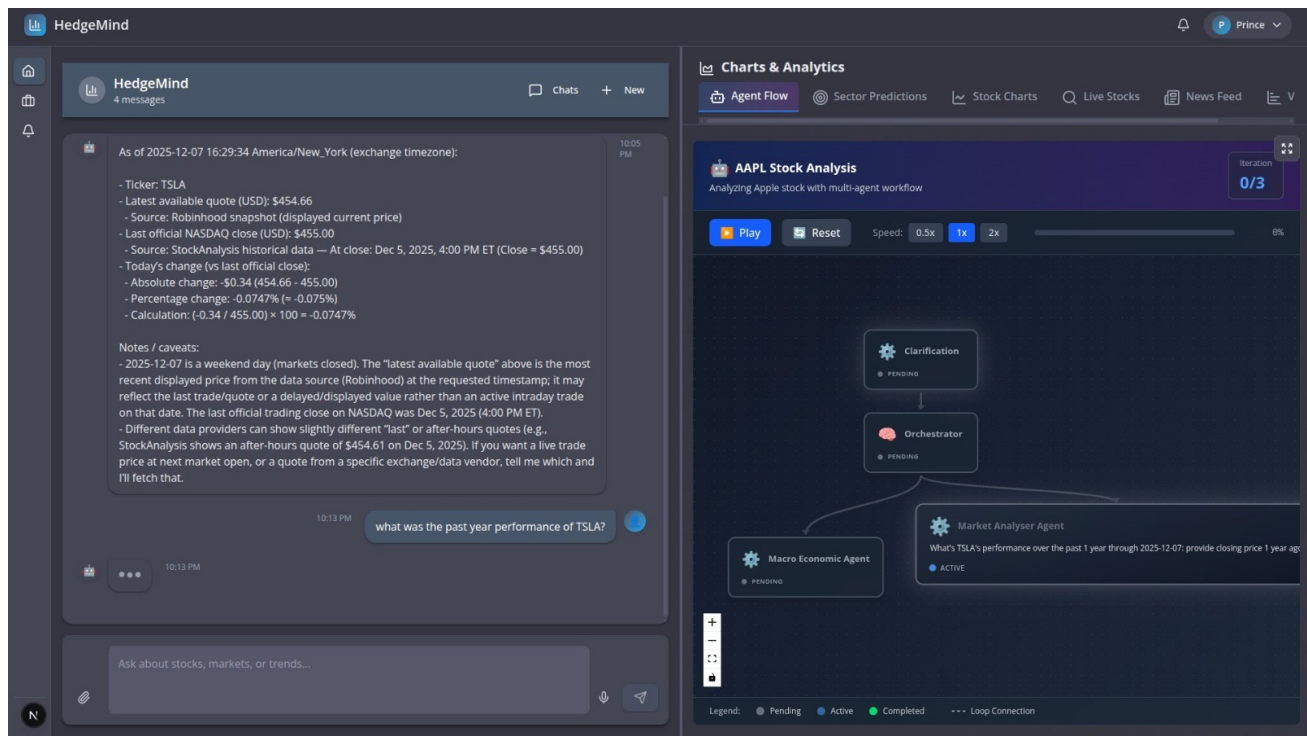
## 5 Challenges Faced and Solutions

- **Asynchronous Multi-User Request Handling:** Adapting request-response patterns to streaming required treating queries as table entries. Asynchronous processing via Pathway UDFs leverages the **Rust-based incremental engine** to maintain constant latency, with **LLM xPack agents** as stateless functions to ensure scalability without conflicts.
- **Streaming-Interaction Synchronization:** Reconciling asynchronous streams with synchronous user queries required decoupling detection from interaction. We implemented an event-driven **Trigger Agent** for continuous background monitoring, bridging the gap between real-time data velocity and on-demand user interaction.
- **Data Acquisition and Quality:** High-frequency data sourcing faced API limits and scarcity. We built custom **WebSocket wrappers** to simulate streaming from historical data, enabling controlled replay for rigorous testing of Pathway's temporal operators.
- **Stateful Multi-Tenancy:** To avoid per-user pipelines, we implemented database-level partitioning by ID. Storing histories and portfolios as streaming tables allows Pathway's operators to route context dynamically, enabling **hori-**

**zontal scaling** and centralized observability.

- **Model Training-Inference Synchronization:** To prevent stale predictions during continuous fine-tuning, we adopted versioned artifacts with timestamped checkpoints. This shared registry implementation uses atomic pointer updates to ensure seamless model synchronization.
- **Deterministic Guardrails against Hallucination:** To mitigate advisory risks, we separated probabilistic reasoning from calculation. The architecture restricts LLMs to strategy formulation while delegating quantitative tasks to **deterministic tools**, ensuring AI suggestions are **mathematically validated**.

## 6 User Interface



**Figure 3: The HedgeMind Advisory Dashboard.** This interface bridges natural language interaction with transparent agentic orchestration. The left panel enables users to query the system (e.g., for real-time stock analysis), while the “Agent Flow” tab on the right visualizes the live decision logic and dynamic routing between the Clarification, Orchestrator, and specialized domain agents. For a complete gallery of UI components and detailed workflow snapshots, please refer to the Appendix.

## 7 Future Work and Conclusion

Our robust production pipeline hybridizes Chronos-2 and SARIMAX (stocks), LSTMs (macro), and GA/GP (strategy). However, architectural traits present non-critical bottlenecks in ultra-high-frequency environments: Transformers suffer from Softmax attention overhead in long contexts, LSTMs face context dilution over extended sequences, and evolutionary algorithms incur high computational costs due to population-based evaluation.

Hence, our next phase focuses on replacing these individual components with a unified **Biologically-Inspired Sparse Architecture (BDH)**. This upgrade will fundamentally transform the pipeline. The BDH architecture utilizes a high-dimensional sparse memory and autocorrelation-based resonance which will allow the system to store distinct, non-overlapping market regimes and effectively act as a native seasonality and cycle detector. BDH implements a gated mechanism where signals are only propagated if they resonate with established memory traces, effectively acting as a noise filter. This ensures that the strategy agent only evolves strategies based on statistically significant signals, drastically reducing compute costs. By deploying this BDH Architecture at full scale, we aim to unify the disparate strengths of our current stack-memory, reasoning, and optimisation-into a single, high-efficiency engine capable of effectively filtering the noise of high-frequency markets.

## References

The bibliography and the accompanying references have been added in the appendix.